The Journal of Open Source Software

Fraxinus: An Open-Source Navigation System for Bronchoscopy Research

Erlend Fagertun Hofstad[1], **Ole Vegard Solberg**[1], **David Bouget**[1], **Thomas Langø**[1,2], **Arne Kildahl-Andersen**[3,4], **Hanne Sorger**[4,5], **Tore Amundsen**[3,4], **and Håkon Olav Leira**[3,4]

1 SINTEF Digital, Department of Health Research, 7030 Trondheim, Norway **2** National Research Center for Minimally Invasive and Image-Guided Diagnostics and Therapy, St. Olavs Hospital, 7030 Trondheim, Norway **3** Department of Thoracic Medicine, St. Olavs Hospital, 7030 Trondheim, Norway **4** Department of Circulation and Medical Imaging, Faculty of Medicine and Health Sciences, Norwegian University of Science and Technology, 7030 Trondheim, Norway **5** Clinic of Medicine, Nord-Trøndelag Hospital Trust, Levanger Hospital, 7601 Levanger, Norway

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 01 September 2026
Published: unpublished

Summary

Fraxinus is an open-source, workflow-driven software system for bronchoscopy planning, developed at SINTEF Digital and St. Olavs hospital, Trondheim, Norway. Built on the CustusX image-guided therapy platform (Askeland et al., 2016), Fraxinus provides a complete pipeline for pre-procedural bronchoscopy planning, with a primary focus on lung cancer staging and diagnosis. Key capabilities include automatic multi-organ segmentation from computed tomography (CT) images using deep learning, positron emission tomography (PET) and CT fusion for metabolically active lymph node visualization, route computation through the airway tree to pulmonary lesions, virtual bronchoscopy, and an endobronchial ultrasound (EBUS) simulator for rehearsing navigated lymph node sampling. The software is written in C++ with Qt for the user interface and runs on Windows and Ubuntu. Fraxinus runs on commodity hardware and supports both CPU and GPU execution allowing it to run on a modern laptop while remaining compatible with GPU acceleration where available. Pre-built installers for Windows and Ubuntu allow clinical researchers to install and run Fraxinus without compilation or programming knowledge.

An earlier version of the Fraxinus architecture was described in (Bakeng et al., 2019). The present paper describes the substantially evolved open-source codebase, which now integrates AI-driven multi-organ segmentation and deformable image registration.

Statement of Need

Lung cancer is the leading cause of cancer-related mortality worldwide, accounting for approximately 1.8 million deaths annually (Bray et al., 2024). Flexible bronchoscopy is the primary diagnostic tool for pulmonary lesions, but the diagnostic yield for peripherally located, bronchoscopically non-visible tumors is as low as 14–20% for lesions smaller than 2 cm in the outer third of the lung (A. Chen et al., 2007; Dhillon & Harris, 2017), compared with around 80% for centrally visible lesions (Rivera et al., 2013). Although CT-guided transthoracic biopsy achieves higher diagnostic yield for small peripheral lesions, bronchoscopic access carries a lower risk of pneumothorax and enables biopsy, staging, and potentially therapeutic procedures within a single endoscopic session. Pre-procedural planning by computing routes through the airway tree to the target lesion and rehearsing the procedure with virtual bronchoscopy (VB) raises the diagnostic yield substantially (Kops et al., 2023).

Open-source, full-pipeline tools for bronchoscopy planning are largely absent from the literature and from public repositories. Fraxinus addresses this gap by providing researchers and clinicians with a freely available research platform that runs on commodity hardware. It is intended for expert personnel in clinical research settings and is not approved for routine clinical use.

State of the Field

Several open-source medical imaging platforms support partial bronchoscopy workflows. 3D Slicer (Fedorov et al., 2012) is widely used for airway segmentation (via extensions such as SlicerAirwaySegmentation) and general virtual endoscopy, but it is a general-purpose framework without a dedicated bronchoscopy workflow. MITK (Wolf et al., 2005) similarly offers broad imaging functionality without a bronchoscopy-specific pipeline.

Commercial bronchoscopy navigation platforms (Medtronic superDimension, Intuitive Ion, J&J Monarch) offer strong mechanical reach through shape-sensing catheters and robotic actuation, with multicenter trials reporting high diagnostic yields for peripheral nodules (Ali et al., 2023; Simoff et al., 2021; TARGET Investigators et al., 2025). However, they require proprietary hardware, cannot be independently inspected or extended, and are inaccessible to most research groups and resource-limited health systems.

Fraxinus is unique in combining components into an open-source package: (1) automated AI-based segmentation of multiple anatomical structures from CT; (2) a guided, step-by-step workflow tailored to bronchoscopy procedure planning; and (3) virtual bronchoscopy visualization.

Software Design

Fraxinus is implemented in C++ using Qt for the GUI, VTK for 3D visualization and image processing. It uses the CTK OSGi plugin framework (Seibert et al., 2010), inherited from CustusX, to separate concerns across three public plugins:

org.custusx.fraxinus provides the application entry point, installer configuration, and setup logic for external AI tools (downloading models, configuring Python virtual environments).

org.custusx.fraxinus.core.state implements a finite state machine that guides users through a sequential workflow: from new patient creation and CT import, through automated segmentation and route planning, to virtual bronchoscopy and procedure planning. Each state configures the application layout, available controls, and automated actions appropriate to that step. This plugin also orchestrates the multi-step segmentation pipeline, invoking Python-based AI tools as subprocesses and importing results back into the patient session.

org.custusx.fraxinus.widgets contains all user-facing interface components. Key widgets handle patient loading, DICOM import, segmented structure visibility, target pinpointing in multi-planar views, virtual bronchoscopy visualization with an integrated EBUS simulator that renders a simulated ultrasound sector from the CT to rehearse navigated lymph node sampling, and procedure planning.

The segmentation pipeline automatically extracts airways, lungs, lung lobes, lymph nodes, heart, major pulmonary vessels, and tumor candidates from a chest CT scan. Two deep learning backends are supported: Raidionics (Bouget et al., 2019, 2022, 2023; Støverud et al., 2024), which provides organ-specific models (CT_Airways, CT_Lungs, CT_LymphNodes, CT_Tumor, and others), and TotalSegmentator (Wasserthal et al., 2023), which can segment over 100 anatomical structures. Deformable image registration via Elastix (Klein et al., 2010; Shamonin et al., 2014) enables fusion of PET and CT volumes for combined metabolic and anatomical visualization.

The Journal of Open Source Software

Once the airway centerline is extracted, a route-to-target algorithm traces the optimal path from the trachea to a clinician-specified lesion through the airway tree. The resulting procedure planning view is shown in Figure 1.

The system is built and distributed via a Python-based superbuild script (`cxFraxinusInstaller.py`) that manages all C++ library dependencies (Qt, VTK, ITK, Eigen, OpenCV, OpenIGTLink, CTK, Boost) and Python tool setup. Pre-built binary installers are provided for Windows and Ubuntu.

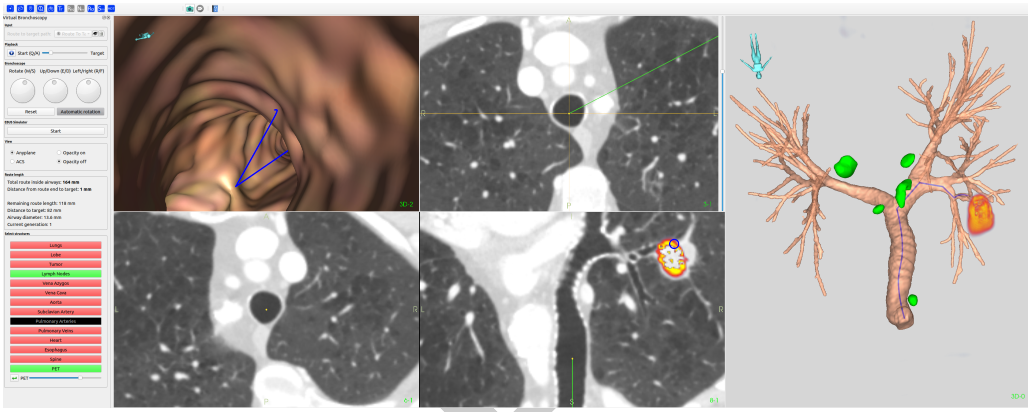

Figure 1: The Fraxinus procedure planning step. Left: Virtual bronchoscopy with the planned route overlaid on the airway surface and 2D CT sections overlaid with PET indicating the target lesion. Right: 3D rendering of the segmented airway tree with lymph nodes (green) and PET-positive lesion (red/yellow).

Research Impact

The Fraxinus development originated from the doctoral thesis of Håkon Olav Leira in 2012 (Leira, 2012), first presented as Fraxinus at IPCAI/CARS 2016 (Heidelberg, Germany) and formally published in 2019 (Bakeng et al., 2019). The system has since been extended with AI-based segmentation and multimodal registration. Fraxinus has been tested and used in clinical research at a total of eight hospitals in Norway, mainly at St. Olavs Hospital (Trondheim, Norway), supporting studies in bronchoscopy planning.

The planning module in Fraxinus has also been included and validated through Sorger et al. (Sorger et al., 2017), who demonstrated feasibility of navigated endobronchial ultrasound (EBUS) bronchoscopy in humans. Kildahl-Andersen et al. (Kildahl-Andersen et al., 2024) subsequently validated PET-CT-fused navigation in a human cohort.

Perhaps the strongest indicator of the platform's maturity is its adoption as the core navigation technology in two independent robotic bronchoscopy platforms. Chen et al. (X.-Y. Chen et al., 2023) developed a teleoperated robotic bronchoscopy system with a variable-stiffness catheter, using CustusX as their multimodal navigation framework. Gruionu et al. (Gruionu et al., 2024) developed the RoboCath robotic catheter platform within the IDEAR project (University of Craiova and SINTEF), using Fraxinus directly for procedural planning and guidance. This adoption demonstrates that the software architecture is sufficiently robust, documented, and generalizable to serve as infrastructure for third-party development.

The software is available at https://github.com/SINTEFMedTek/Fraxinus under the BSD 3-Clause License. Binary installers and sample phantom datasets are provided at https://custusx.pages.sintef.no/fraxinus/. The codebase serves as a reference implementation for bronchoscopy planning research.

AI Usage Disclosure

Claude Sonnet 4.6 (Anthropic) was used to assist with drafting of manuscript text and compilation of the reference list. All AI-assisted content has been reviewed, edited, and validated by the human authors, who take full responsibility for the accuracy and completeness of the paper.

Acknowledgements

The authors thank the patients, clinical staff, and research colleagues at the Department of Thoracic Medicine, St. Olavs hospital, Trondheim, and at SINTEF Digital, Department of Health Research, who have contributed to Fraxinus with procedural expertise, software development and iterative testing. Development of Fraxinus has been supported by the Norwegian National Research Center for Minimally Invasive and Image-Guided Diagnostics and Therapy (MiDT) at Center for Innovation, Medical Devices and Technology (SIMUT) at St. Olavs hospital, SINTEF Digital, the Norwegian Research Council, the Norwegian Cancer Society, the Liaison Committee between Central Norway Regional Health Authority and NTNU, and the EEA project IDEAR.

References

Ali, M. S., Trick, W., Mba, B. I., & others. (2023). Diagnostic performance and safety profile of robotic-assisted bronchoscopy: A systematic review and meta-analysis. *Annals of the American Thoracic Society*, *20*(11), 1555–1564. https://doi.org/10.1513/AnnalsATS.202211-940SR

Askeland, C., Solberg, O. V., Bakeng, J. B. L., Reinertsen, I., Tangen, G. A., Hofstad, E. F., Iversen, D. H., Våpenstad, C., Selbekk, T., Langø, T., Hernes, T. A. N., Elvestuen, H., Leira, H. O., & Lindseth, F. (2016). CustusX: An open-source research platform for image-guided therapy. *International Journal of Computer Assisted Radiology and Surgery*, *11*(4), 505–519. https://doi.org/10.1007/s11548-015-1292-0

Bakeng, J. B. L., Vegge, L. C. N., Askeland, C., Hofstad, E. F., Cael, T., Wastvedt, S. L., Husabø, O., Tangen, G. A., Leira, H. O., & Langø, T. (2019). Using the CustusX toolkit to create an image guided bronchoscopy application: Fraxinus. *PLOS ONE*, *14*(2), e0211772. https://doi.org/10.1371/journal.pone.0211772

Bouget, D., Alsinan, D., Gaitan, V., Holden Helland, R., Pedersen, A., Solheim, O., & Reinertsen, I. (2023). Raidionics: An open software for pre- and postoperative central nervous system tumor segmentation and standardized reporting. *Scientific Reports*, *13*, 14921. https://doi.org/10.1038/s41598-023-42048-7

Bouget, D., Jørgensen, A., Kiss, G., Leira, H. O., & Langø, T. (2019). Semantic segmentation and detection of mediastinal lymph nodes and anatomical structures in CT data for lung cancer staging. *International Journal of Computer Assisted Radiology and Surgery*, *14*, 977–986. https://doi.org/10.1007/s11548-019-01948-8

Bouget, D., Pedersen, A., Vanel, J., Leira, H. O., & Langø, T. (2022). Mediastinal lymph nodes segmentation using 3D convolutional neural network ensembles and anatomical priors guiding. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*. https://doi.org/10.1080/21681163.2022.2043778

Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R. L., Soerjomataram, I., & Jemal, A. (2024). Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, *74*(3), 229–263. https://doi.org/10.3322/caac.21834

Chen, A., Pastis, N., Furukawa, B., & Silvestri, G. A. (2007). The effect of physician specialty on bronchoscopy use and diagnostic yield for lung cancer. *Chest*, *131*(6), 1806–1811. https://doi.org/10.1378/chest.06-2982

Chen, X.-Y., Xiong, X., Wang, X., Li, P., Wang, S., Akinyemi, T., Duan, W., Du, W., Omisore, O., & Wang, L. (2023). *Design of a teleoperated robotic bronchoscopy system for peripheral pulmonary lesion biopsy*. https://doi.org/10.48550/arXiv.2306.09598

Dhillon, S. S., & Harris, K. (2017). Bronchoscopy for the diagnosis of peripheral lung lesions. *Journal of Thoracic Disease*, *9*(S10), S1047–S1058. https://doi.org/10.21037/jtd.2017.05.48

Fedorov, A., Beichel, R., Kalpathy-Cramer, J., Finet, J., Fillion-Robin, J.-C., Pujol, S., Bauer, C., Jennings, D., Fennessy, F., Sonka, M., & others. (2012). 3D Slicer as an image computing platform for the quantitative imaging network. *Magnetic Resonance Imaging*, *30*(9), 1323–1341. https://doi.org/10.1016/j.mri.2012.05.001

Gruionu, L. G., Langø, T., Leira, H. O., Hofstad, E. F., Udriștoiu, A. L., Iacob, A. V., Constantinescu, C., Chihaia, C., & Gruionu, G. (2024). Design and technical feasibility testing of a medical robot for flexible catheter navigation inside the lung airways. *bioRxiv*. https://doi.org/10.1101/2024.05.01.592024

Kildahl-Andersen, A., Hofstad, E. F., Solberg, O. V., Sorger, H., Amundsen, T., Langø, T., & Leira, H. O. (2024). Navigated ultrasound bronchoscopy with integrated positron emission tomography — a human feasibility study. *PLOS ONE*, *19*(8), e0305785. https://doi.org/10.1371/journal.pone.0305785

Klein, S., Staring, M., Murphy, K., Viergever, M. A., & Pluim, J. P. W. (2010). Elastix: A toolbox for intensity-based medical image registration. *IEEE Transactions on Medical Imaging*, *29*(1), 196–205. https://doi.org/10.1109/TMI.2009.2035616

Kops, S. E. P., Heus, P., Korevaar, D. A., & others. (2023). Diagnostic yield and safety of navigation bronchoscopy: A systematic review and meta-analysis. *Lung Cancer*, *180*, 107196. https://doi.org/10.1016/j.lungcan.2023.107196

Leira, H. O. (2012). *Development of an image guidance research system for bronchoscopy* [PhD thesis]. Norwegian University of Science; Technology.

Rivera, M. P., Mehta, A. C., & Wahidi, M. M. (2013). Establishing the diagnosis of lung cancer: Diagnosis and management of lung cancer, 3rd ed: American college of chest physicians evidence-based clinical practice guidelines. *Chest*, *143*(5), e142S–e165S. https://doi.org/10.1378/chest.12-2353

Seibert, D., Stein, D., Burg, K., Blezek, D., Finet, J., Yarmarkovich, A., Rankin, A., Liu, Y., Zuber, M., & Herr, D. (2010). The Common Toolkit (CTK). *Medical Imaging 2010: Advanced PACS-Based Imaging Informatics and Therapeutic Applications*, *7628*, 762811. https://doi.org/10.1117/12.843786

Shamonin, D. P., Bron, E. E., Lelieveldt, B. P. F., Smits, M., Klein, S., & Staring, M. (2014). Fast parallel image registration on CPU and GPU for diagnostic classification of Alzheimer's disease. *Frontiers in Neuroinformatics*, *7*, 50. https://doi.org/10.3389/fninf.2013.00050

Simoff, M. J., Pritchett, M. A., Saltos, A., & others. (2021). Shape-sensing robotic-assisted bronchoscopy for pulmonary nodules: Initial multicenter experience using the Ion endoluminal system. *BMC Pulmonary Medicine*, *21*(1), 322. https://doi.org/10.1186/s12890-021-01671-8

Sorger, H., Hofstad, E. F., Amundsen, T., Langø, T., Bakeng, J. B. L., & Leira, H. O. (2017). A multimodal image guiding system for navigated ultrasound bronchoscopy (EBUS): A human feasibility study. *PLOS ONE*, *12*(2), e0171841. https://doi.org/10.1371/journal.pone.0171841

Støverud, K.-H., Bouget, D., Pedersen, A., Leira, H. O., Amundsen, T., Langø, T., & Hofstad, E. F. (2024). AeroPath: An airway segmentation benchmark dataset with challenging pathology and baseline method. *PLOS ONE*, *19*(10), e0311416. https://doi.org/10.1371/journal.pone.0311416

TARGET Investigators, Murgu, S., Chen, A. C., Gilbert, C. R., Sterman, D. H., Pederson, D., Rafeq, S., Laxmanan, B., Schwiers, M. L., Connelly, J., Benz, H. L., Yasufuku, K., & Silvestri, G. A. (2025). A prospective, multicenter evaluation of safety and diagnostic outcomes with robotic-assisted bronchoscopy: Results of the transbronchial biopsy assisted by robot guidance in the evaluation of tumors of the lung (TARGET) trial. *Chest*, *168*(2), 539–555. https://doi.org/10.1016/j.chest.2025.04.022

Wasserthal, J., Breit, H.-C., Meyer, M. T., Pradella, M., Hinck, D., Sauter, A. W., Heye, T., Boll, D. T., Cyriac, J., Yang, S., Bach, M., & Segeroth, M. (2023). TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, *5*(5), e230024. https://doi.org/10.1148/ryai.230024

Wolf, I., Vetter, M., Wegner, I., Böttger, T., Nolden, M., Schöbinger, M., Hastenteufel, M., Kunert, T., & Meinzer, H.-P. (2005). The medical imaging interaction toolkit. *Medical Image Analysis*, *9*(6), 594–604. https://doi.org/10.1016/j.media.2005.04.005